

IT3010 – Network Design & Management

Creating Virtual Machines and Network Configuration

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- Practical Number: 01

Introduction

The core focus of this practical exercise is to implement virtualization as a method to run multiple, isolated operating systems on a single physical host. By leveraging a Type-2 Hypervisor, we can effectively manage system resources like CPU and RAM while creating an independent network infrastructure. In this task, a CentOS-based server and a Fedora-based workstation were deployed to simulate a real-world client-server networking scenario.

In this practical, two virtual machines were created using a virtualization platform. CentOS Stream 9 was used as the server operating system, and Fedora Workstation 39 was used as the client operating system. Both machines were connected using a virtual network, and static IP addressing was configured using command line tools.

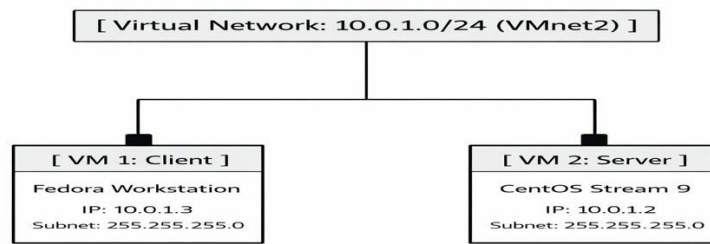
Objectives

The objectives of this practical are:

- To explore the practical implementation of hardware virtualization
- To create virtual machines using a virtualization platform
- To create and configure a virtual network
- To manage and identify network interfaces through the Command Line Interface (CLI).
- To verify cross-node connectivity using network troubleshooting tools (Ping)

NDM Domain Diagram

The following diagram illustrates the network topology used in this practical. Both the server and client virtual machines are connected to the same virtual network.



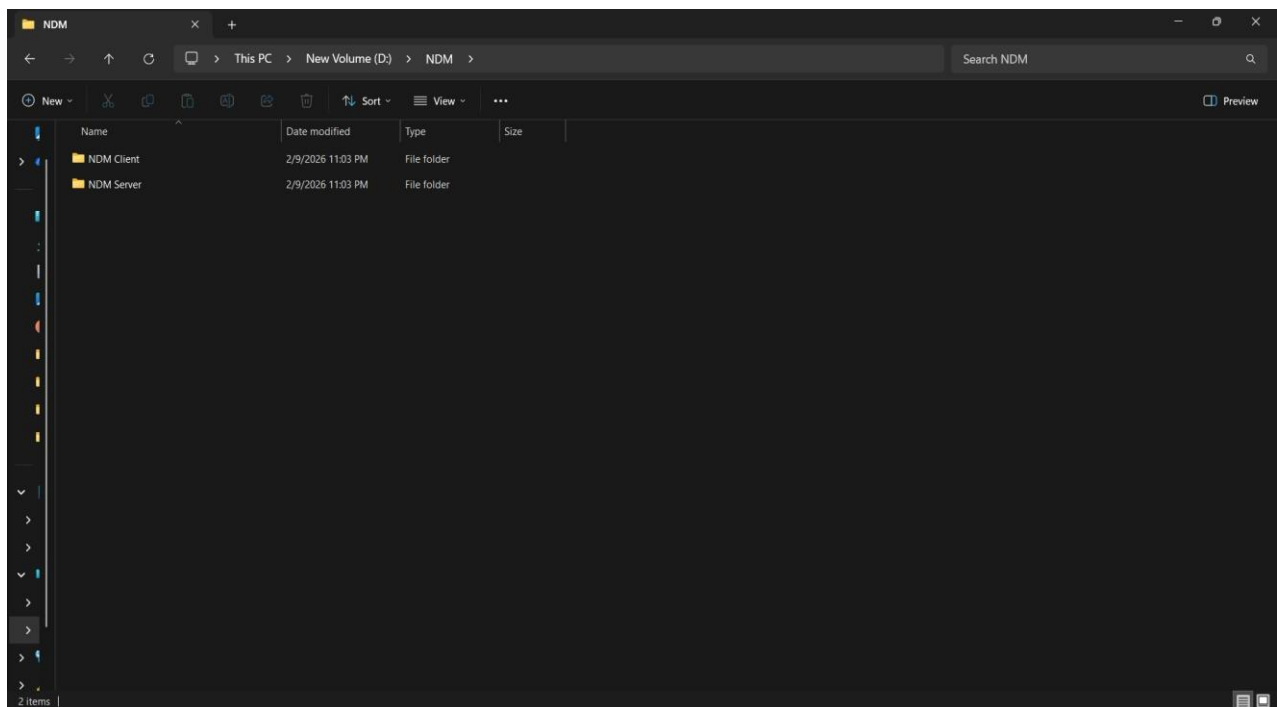
Creating Directory Structure

A directory structure was created in the E drive of the host machine to organize all virtual machine related files.

The following directories were created:

- NDM
- NDM-Server
- NDM-Client

These directories were used to store the server and client virtual machine files separately.

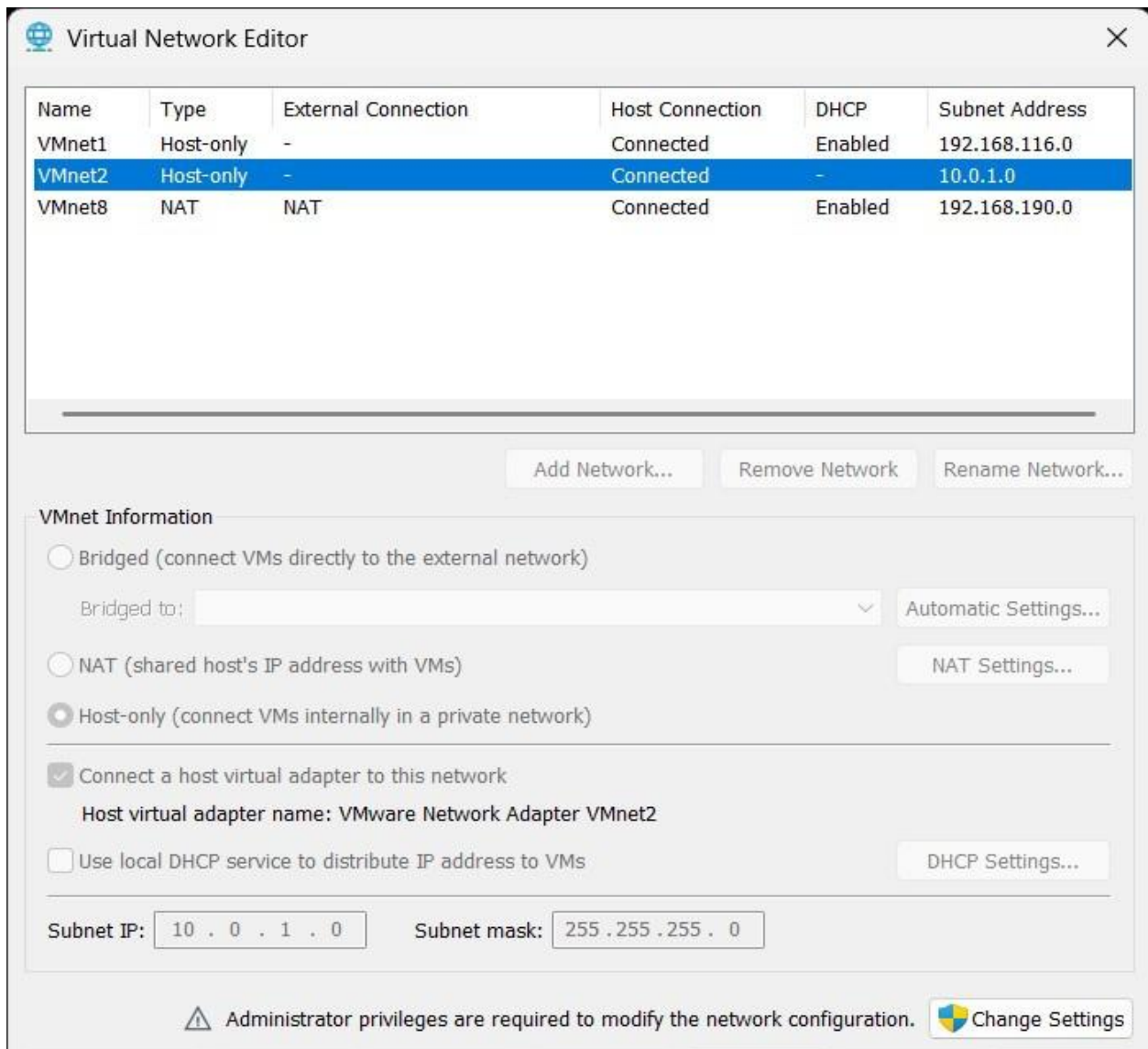


Creating Virtual Network (VMnet2)

A virtual network named VMnet2 was created using the Virtual Network Editor of the virtualization platform. This network allows communication between the CentOS server and Fedora client.

The following network settings were configured:

- Subnet IP: 10.0.1.0
- Subnet Mask: 255.255.255.0
- Network Type: Host only



Creating CentOS Stream 9 Server Virtual Machine

A CentOS Stream 9 virtual machine was created to function as the server. The server was installed without a graphical user interface.

Server configuration details:

- VM Name: Server
- Operating System: CentOS Stream 9 (64-bit)
- RAM: 1.5 GB
- Hard Disk: 15 GB
- Processors: 1
- Root Password: root

- Username/Password: svr1 / svr1
- Network Adapters: VMnet2 and NAT

New Virtual Machine Wizard

✕

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐

Installer disc:

No drives available

☒

Installer disc image file (iso):

D:\CentOS-Stream-9-latest-x86_64-dvd1.iso

Browse...

 CentOS version 5 and earlier 64-bit detected.
This operating system will use Easy Install. [\(What's this?\)](#)

☐

I will install the operating system later.

The virtual machine will be created with a blank hard disk.

New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Server

Location:

D:\NDM\NDM Server

Browse...

The default location can be changed at Edit > Preferences.

New Virtual Machine Wizard

×

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

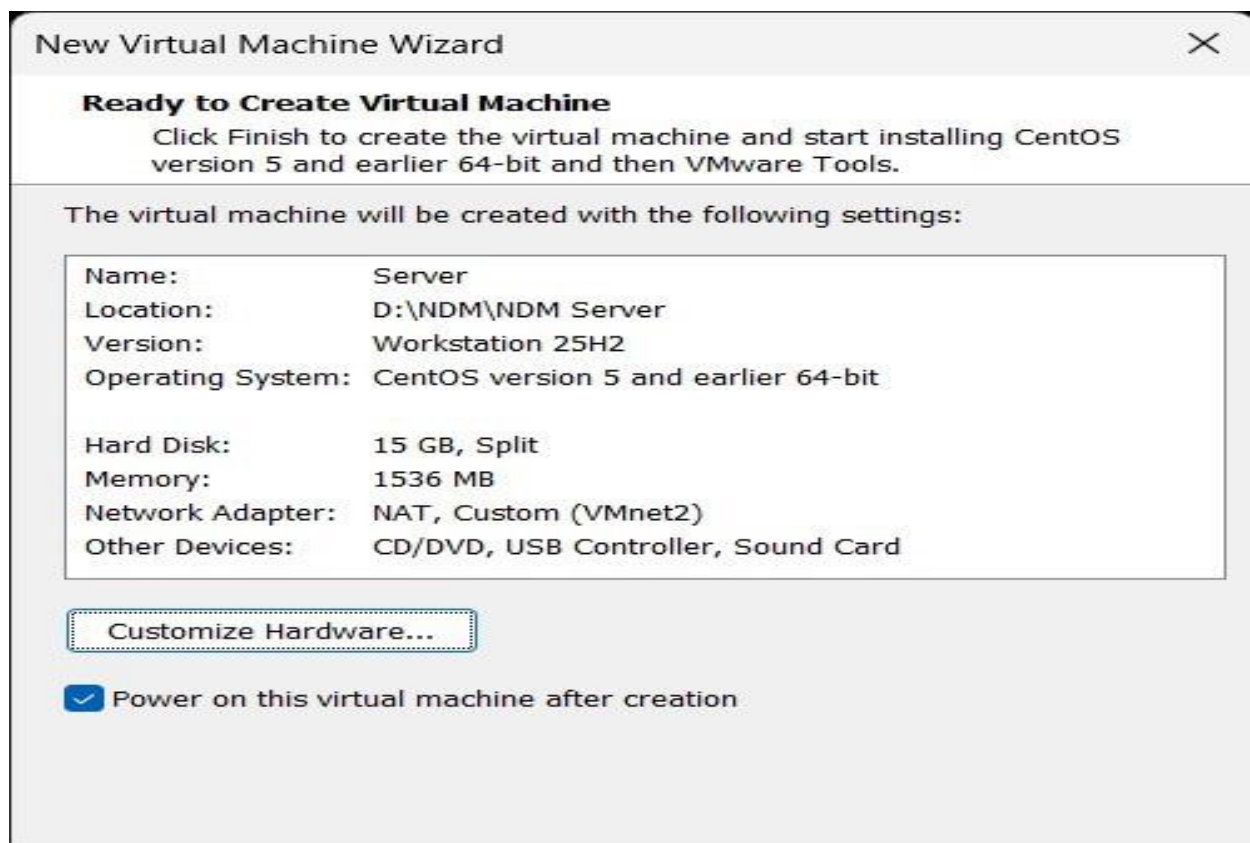
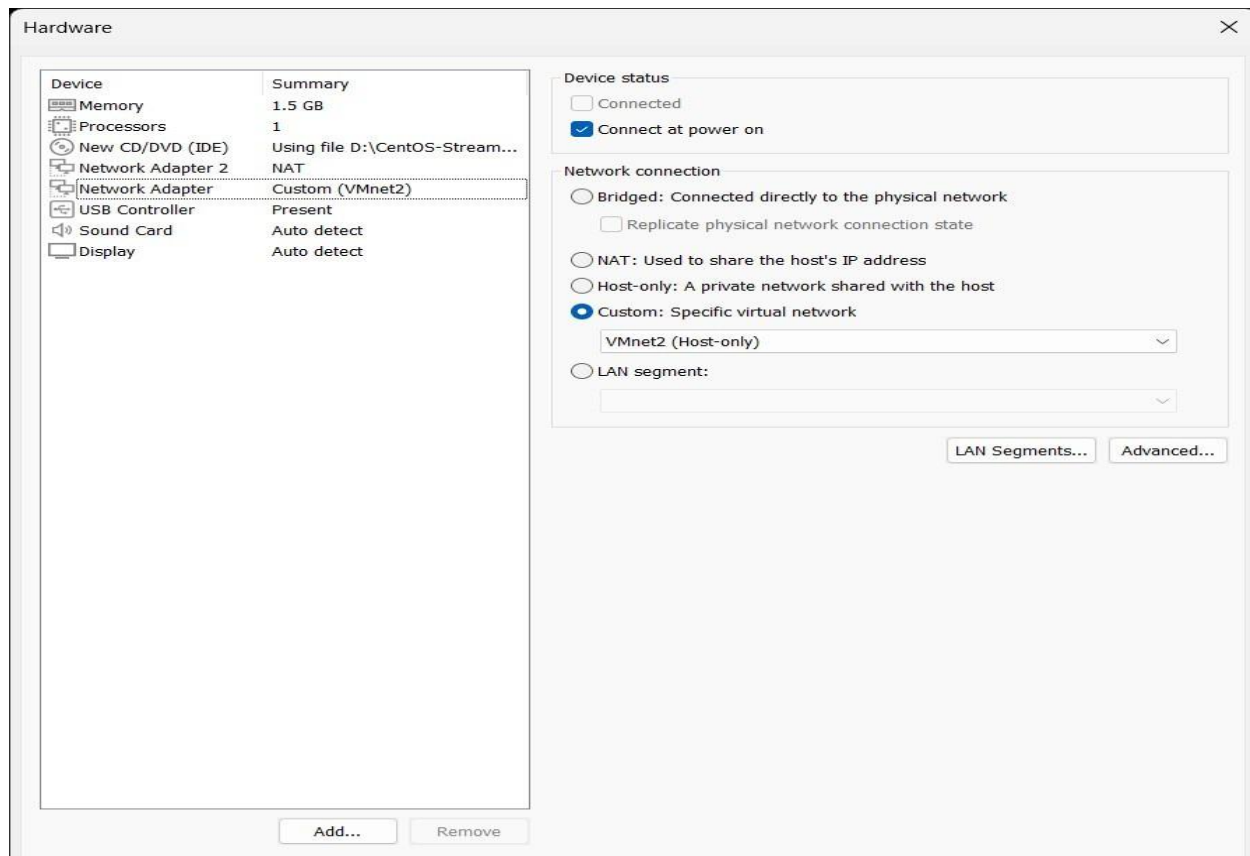
Maximum disk size (GB):

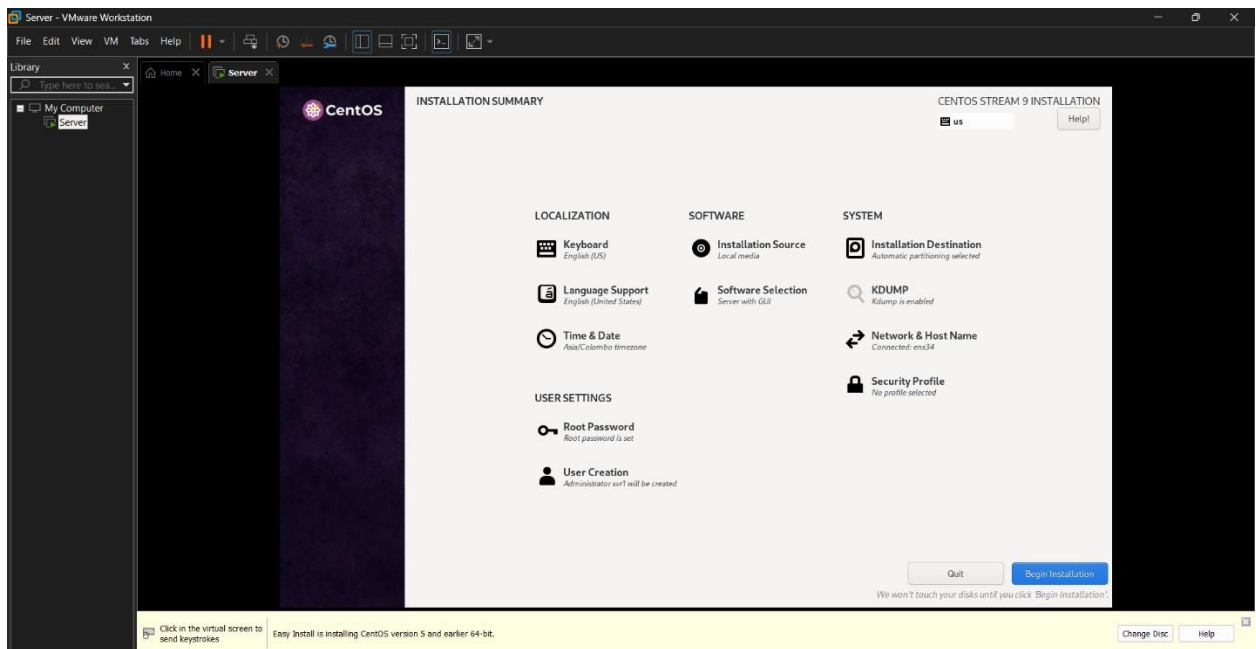
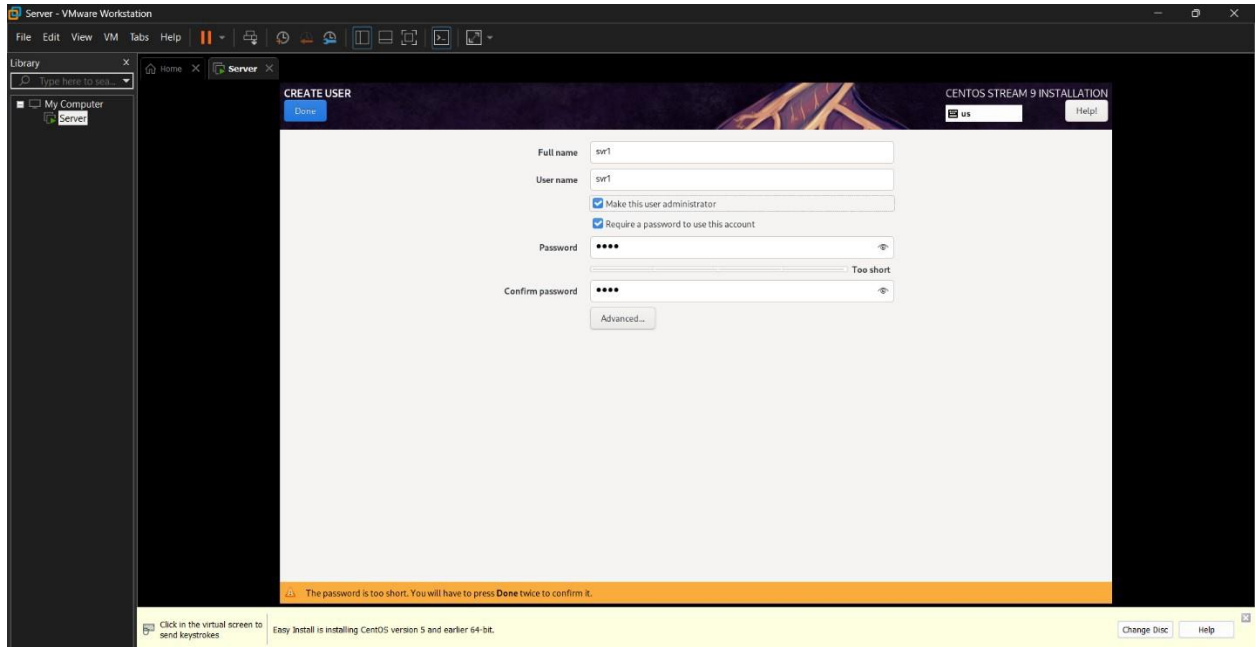
Recommended size for CentOS version 5 and earlier 64-bit: 20 GB

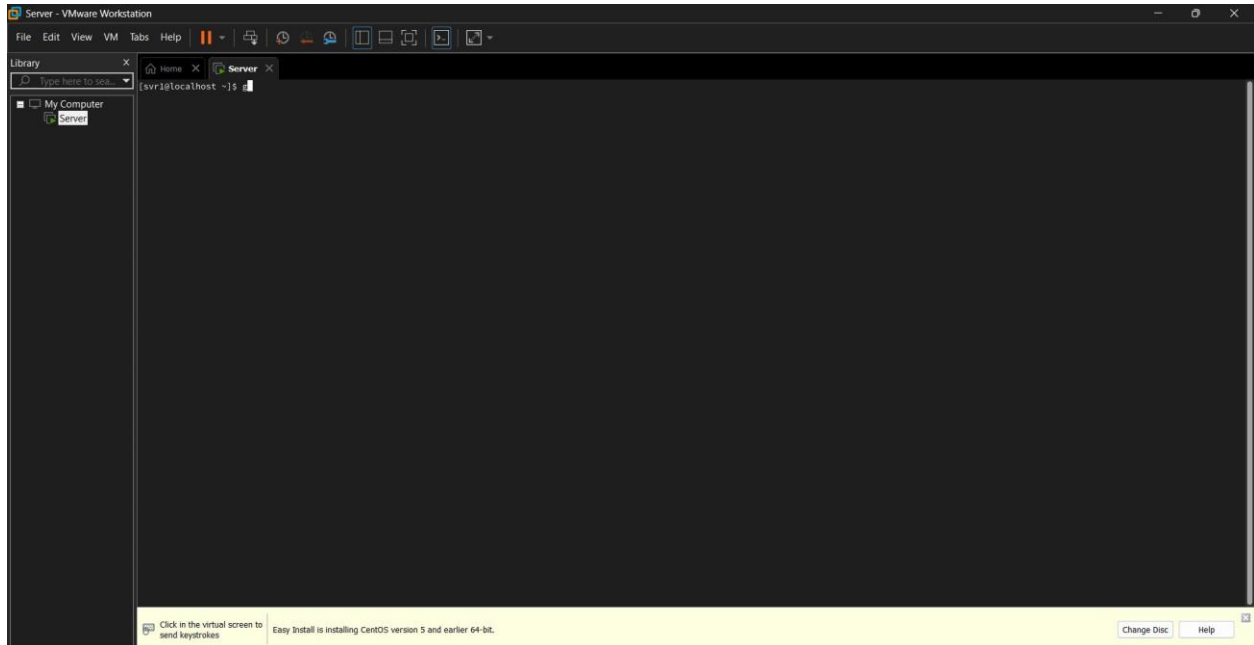
☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.







Creating Fedora Workstation 39 Client Virtual Machine

A Fedora Workstation 39 virtual machine was created as the client workstation.

Client configuration details:

- VM Name: Client
- Operating System: Fedora Workstation 39 (64-bit)
- RAM: 1.5 GB
- Hard Disk: 15 GB
- Processors: 1
- Root Password: root
- Username/Password: client1 / client1
- Network Adapters: VMnet2 and NAT

New Virtual Machine Wizard

✕

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☒ Installer disc image file (iso):

D:\Fedora-Workstation-Live-x86_64-39-1.5 (1).iso

Browse...

☐ I will install the operating system later.

The virtual machine will be created with a blank hard disk.

 Fedora 64-bit detected.
To use Easy Install, insert the first disc of the set.

New Virtual Machine Wizard

✕

Name the Virtual Machine
What name would you like to use for this virtual machine?

Virtual machine name:

Client

Location:

D:\NDM\NDM Client

Browse...

The default location can be changed at Edit > Preferences.

New Virtual Machine Wizard

Specify Disk Capacity

How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB):

Recommended size for Fedora 64-bit: 20 GB

☐ Store virtual disk as a single file

☒ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

Hardware

Device	Summary
Memory	1.5 GB
Processors	1
New CD/DVD (SATA)	Using file D:\Fedora-Workst...
Network Adapter	Custom (VMnet2)
Network Adapter 2	NAT
USB Controller	Present
Sound Card	Auto detect
Display	Auto detect

Add...

Remove

Device status

☐ Connected

☒ Connect at power on

Network connection

☐ Bridged: Connected directly to the physical network

☐ Replicate physical network connection state

☒ NAT: Used to share the host's IP address

☐ Host-only: A private network shared with the host

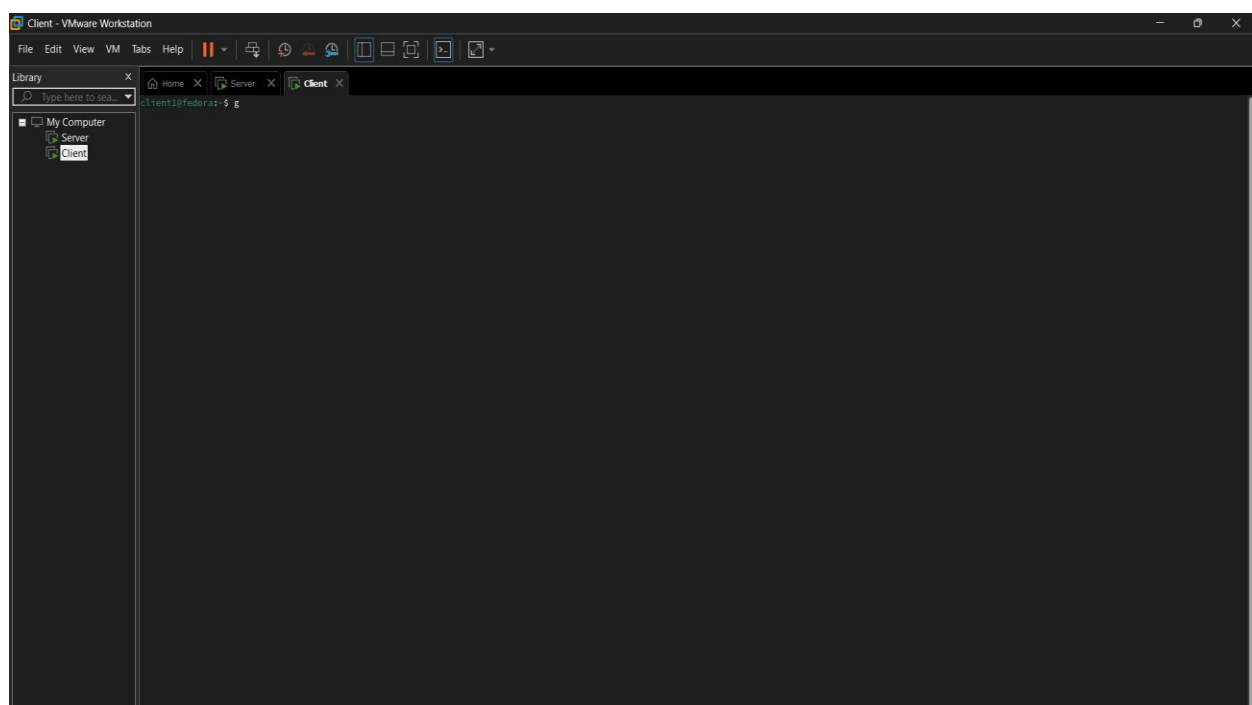
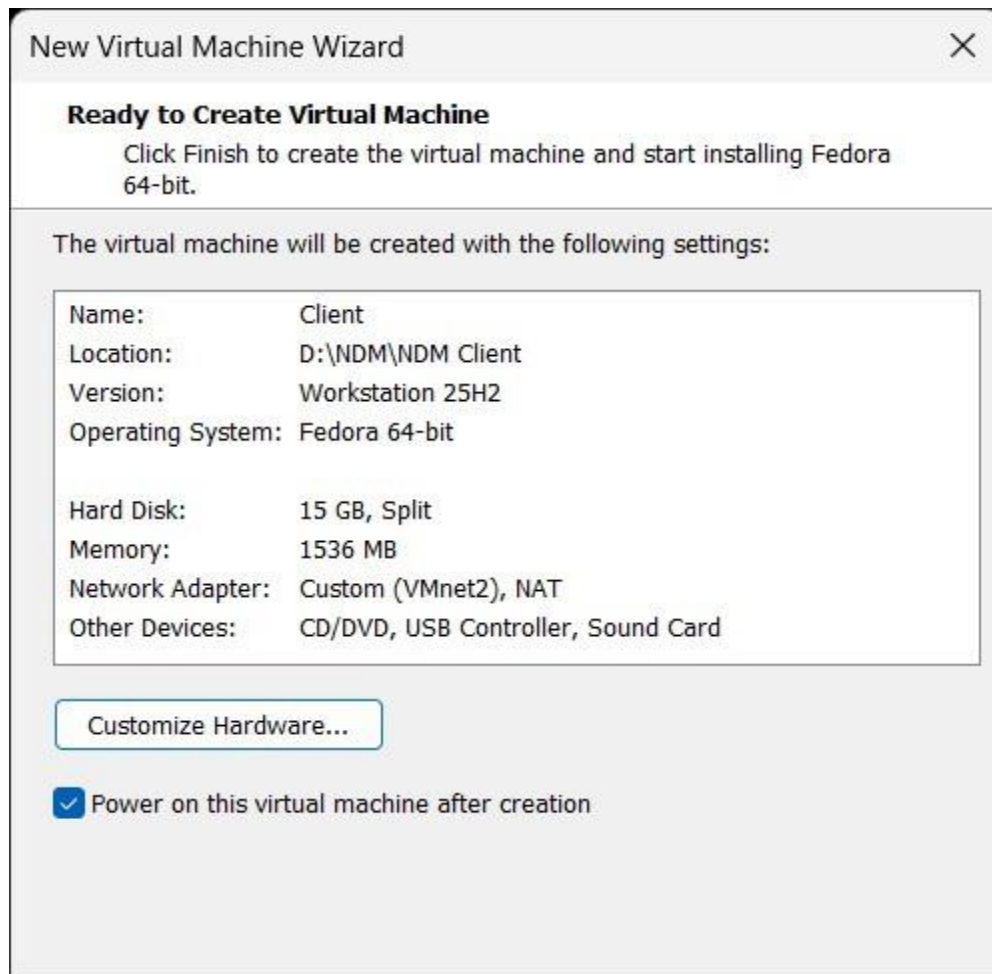
☐ Custom: Specific virtual network

VMnet0

☐ LAN segment:

LAN Segments...

Advanced...



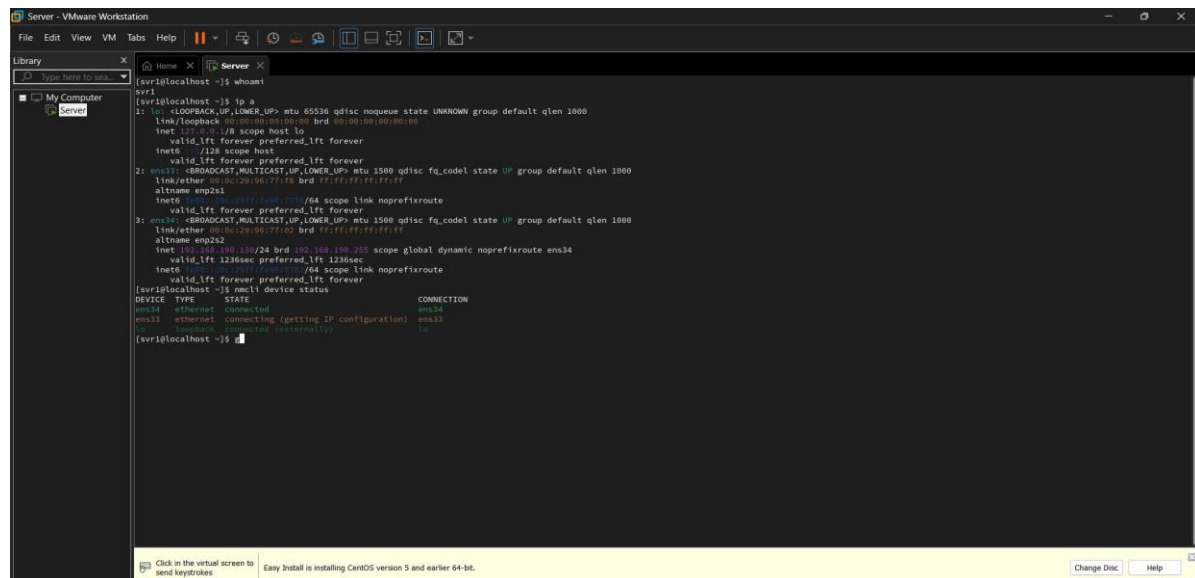
Identifying Network Adapters Using CLI

The ip a command was executed on both the server and client machines to identify the available network adapters.

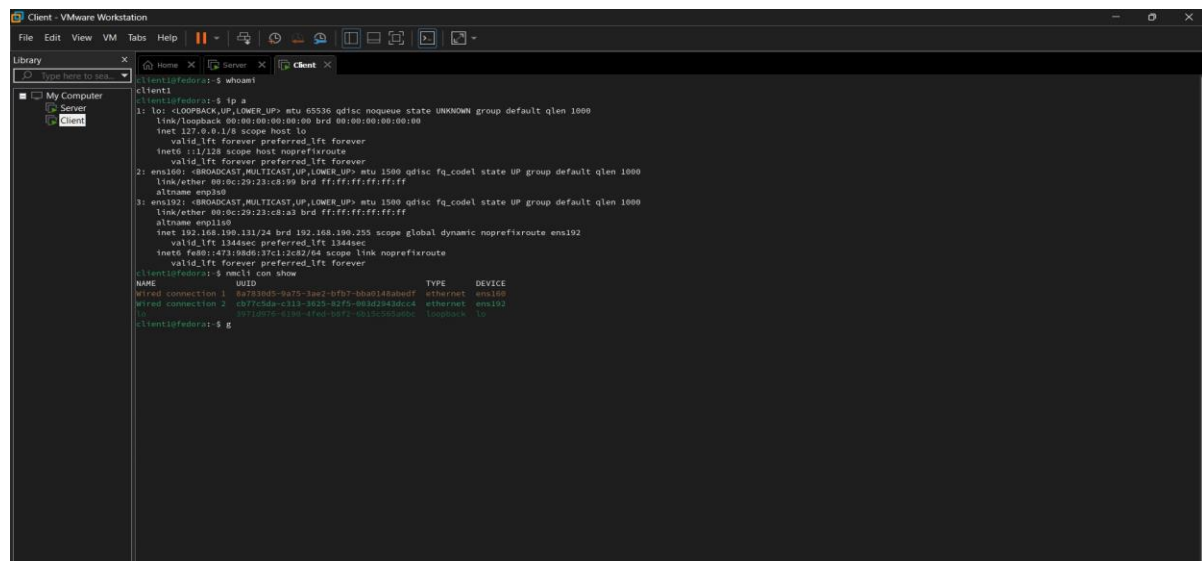
The adapter connected to NAT was identified by an IP address in the 192.168.x.x range, which is automatically assigned by the virtualization platform.

The adapter connected to VMnet2 was identified for manual configuration and assigned a static IP address as required in the practical.

This step was necessary to correctly configure the network settings and ensure communication between the server and client machines.



```
[server@localhost ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host 
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:2b:00:12:13 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.100.13/24 brd 192.168.100.255 scope global dynamic noprefroute ens33
        valid_lft 1336sec preferred_lft 1336sec
    inet6 fe80::2b00:1213:1213:1213/64 scope link noprefroute
        valid_lft forever preferred_lft forever
[server@localhost ~]$ nmcli device status
DEVICE  TYPE      STATE      CONNECTION
ens33    ethernet  connected  ens33
ens33    ethernet  connecting (getting IP configuration)  ens33
lo       loopback  connected  lo
[server@localhost ~]$
```



```
client@fedora:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host 
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3c:12:13 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3c:12:13 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
    inet 192.168.100.13/24 brd 192.168.100.255 scope global dynamic noprefroute ens192
        valid_lft 1344sec preferred_lft 1344sec
    inet6 fe80::273c:1213:1213:1213/64 scope link noprefroute
        valid_lft forever preferred_lft forever
client@fedora:~$ nmcli con show
NAME                                UUID                                  TYPE      DEVICE
wired-connection-1                 8a7810d5-9a75-3ae2-b7d7-6b0d148a8d0f  ethernet  ens160
wired-connection-2                 c877c5da-c313-3623-82f5-083d2943dc04  ethernet  ens192
wired-connection-3                 07120718-8200-474d-9077-0a13c50a0a0c  loopback   lo
client@fedora:~$
```

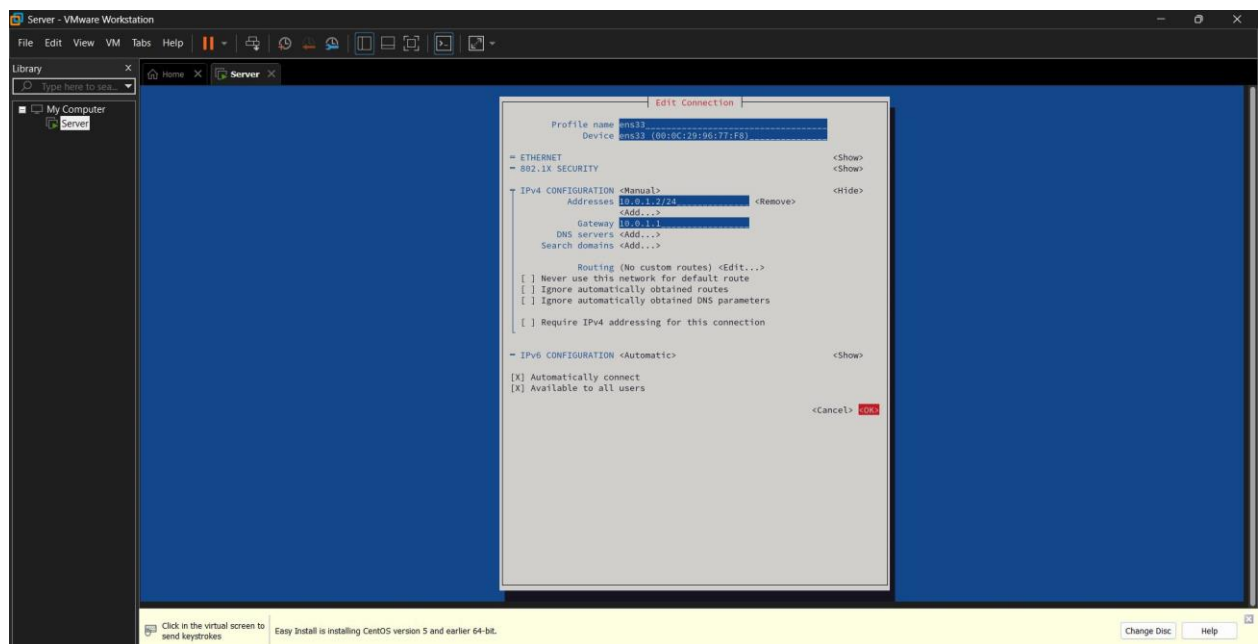
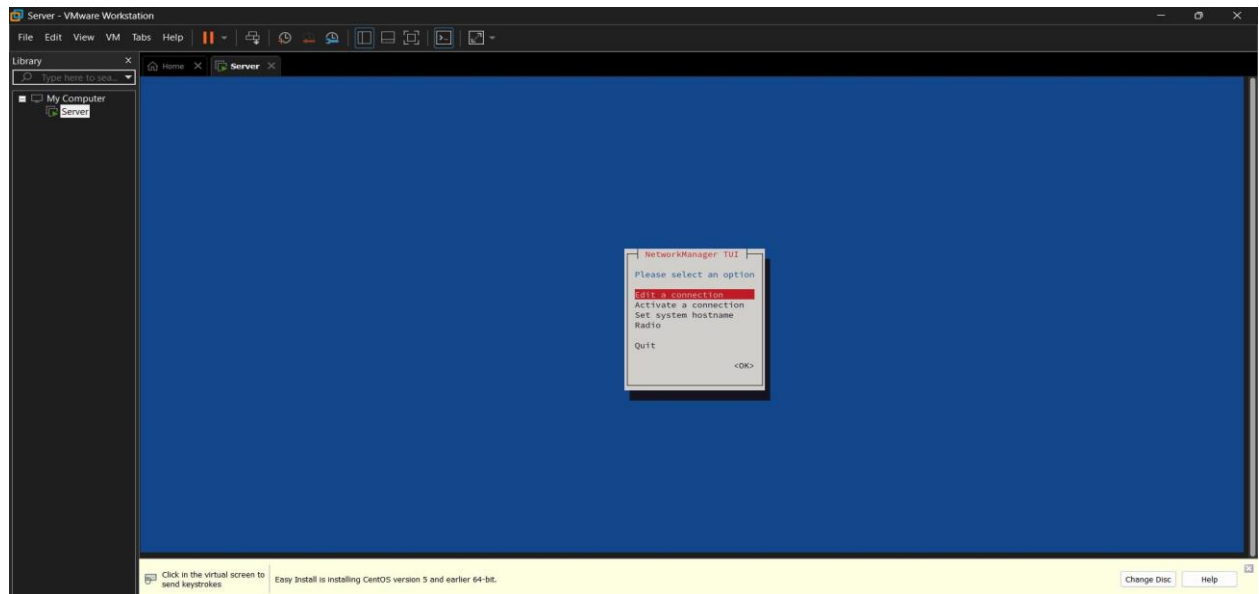
Server Network Configuration (CentOS Stream 9)

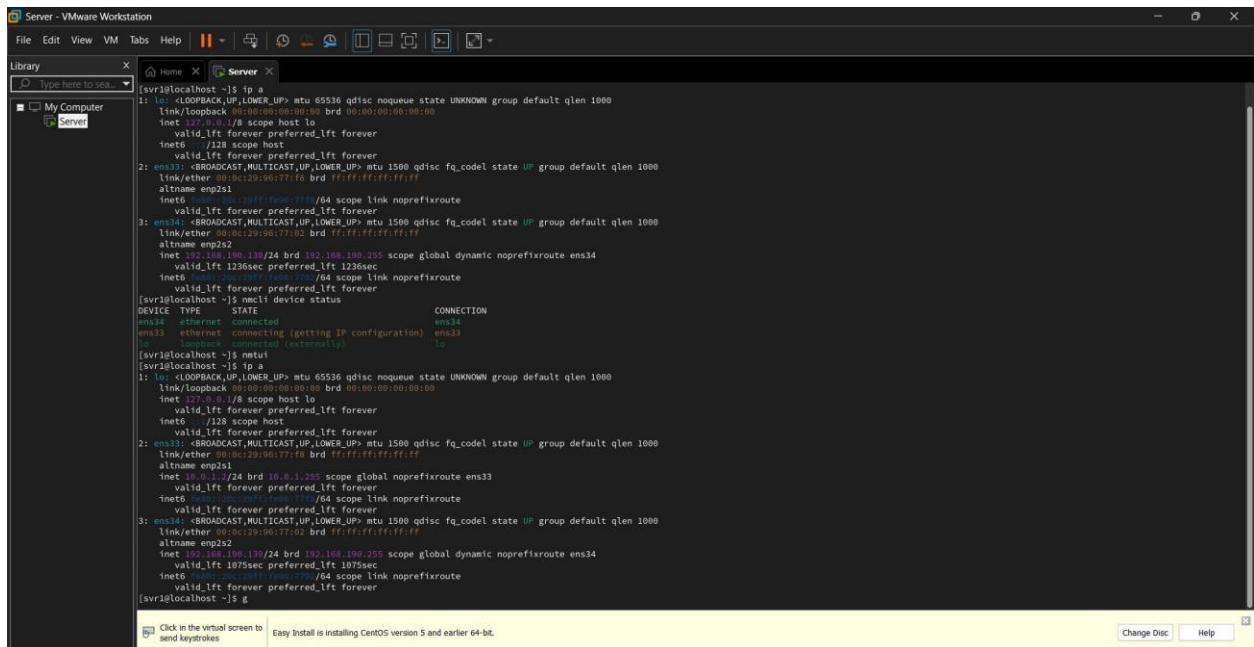
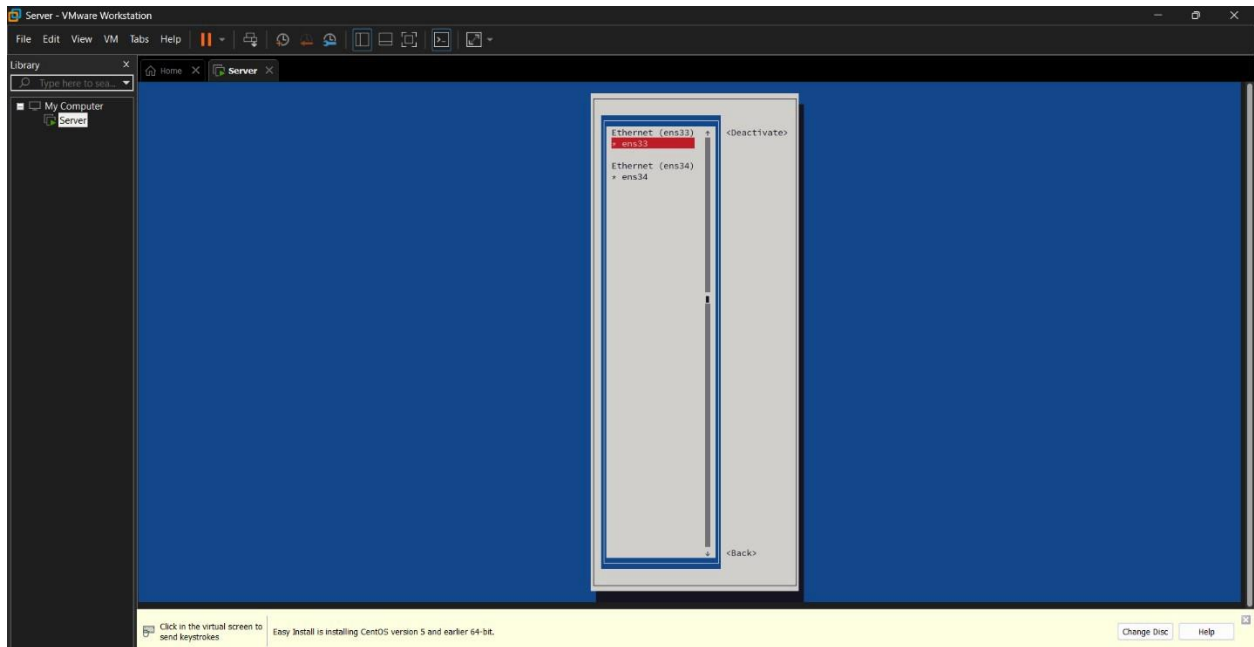
The CentOS server network configuration was performed using the `nmtui` command.

The following static IP configuration was applied to the VMnet2 adapter:

- IP Address: 10.0.1.2/24
- Gateway: 10.0.1.1

The network connection was activated after saving the configuration.





Client Network Configuration Using CLI (Fedora 39)

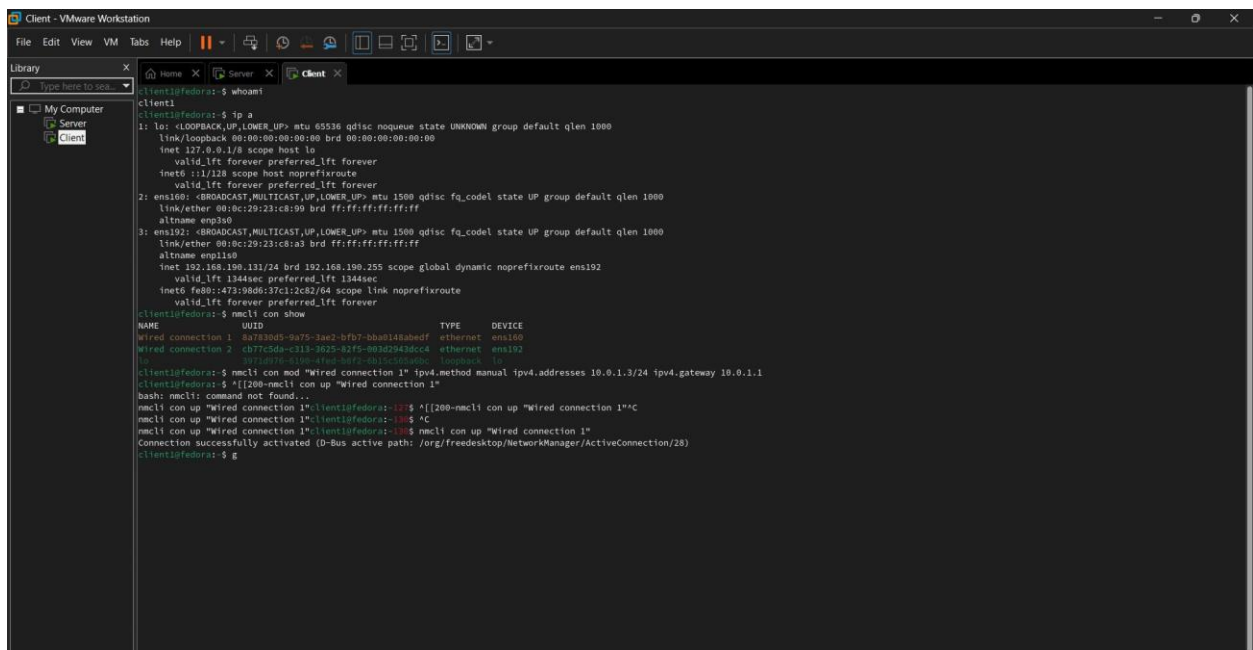
The Fedora client network configuration was completed using command line interface only, as required by the lab instructions.

The following steps were performed:

- Listing network devices using CLI
- Selecting the VMnet2 adapter
- Assigning a static IP address
- Activating the network connection

The following IP configuration was applied:

- IP Address: 10.0.1.3/24
- Gateway: 10.0.1.1

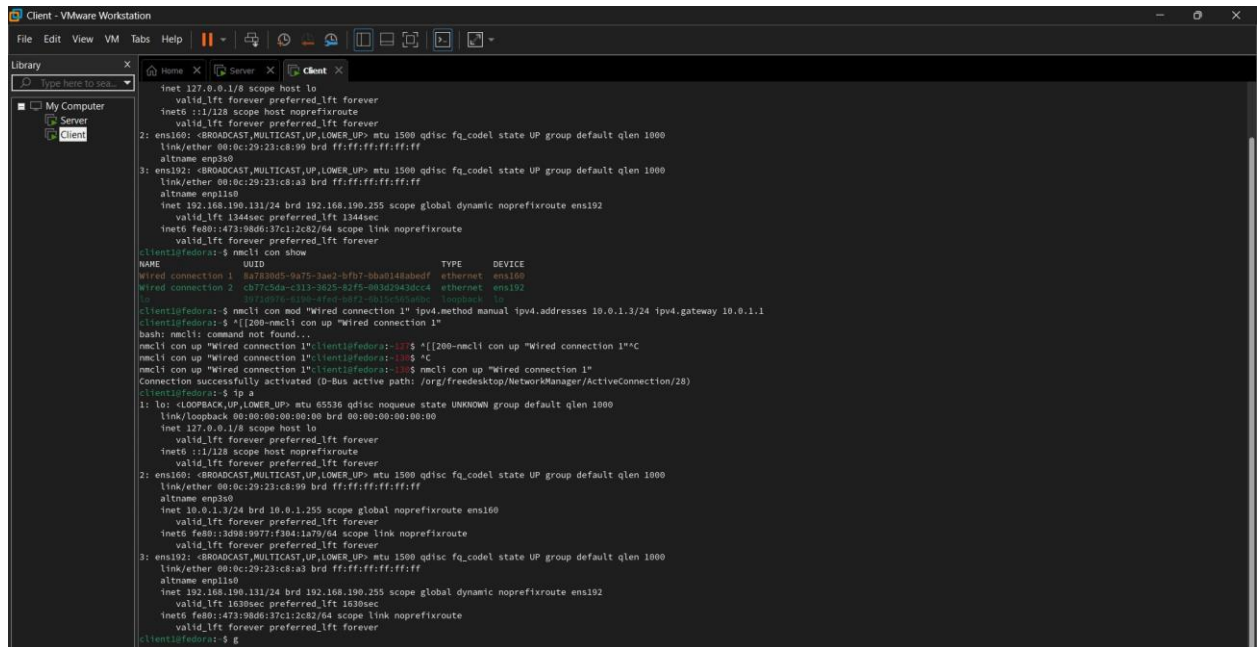


```
Client - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Server
Client

client@fedora:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:23:c8:99 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
    altname enp11s0
    inet 192.168.190.131/24 brd 192.168.190.255 scope global dynamic noprefixroute ens192
        valid_lft 1344sec preferred_lft 1344sec
    inet6 fe80::473:9806:37c1:2c82/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

client@fedora:~$ nmcli con show
NAME                UUID                                  TYPE      DEVICE
wired connection 1  8a7828d5-9a75-3ae2-bf97-8b0d1d8abedf ethernet ens160
wired connection 2  4b77c5da-c313-2625-82f5-003d2d43dcca ethernet ens192
lo                  39712970-6100-4f60-8075-0015c5008000 loopback  lo

client@fedora:~$ nmcli con mod "wired connection 1" ipv4.method manual ipv4.addresses 10.0.1.3/24 ipv4.gateway 10.0.1.1
client@fedora:~$ nmcli con up "wired connection 1"
bash: nmcli: command not found...
nmcli con up "wired connection 1" client@fedora:~$ [[200-nmcli con up "wired connection 1"]C
nmcli con up "wired connection 1" client@fedora:~$ [[185-C
nmcli con up "wired connection 1" client@fedora:~$ [[185-nmcli con up "wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/28)
client@fedora:~$ g
```



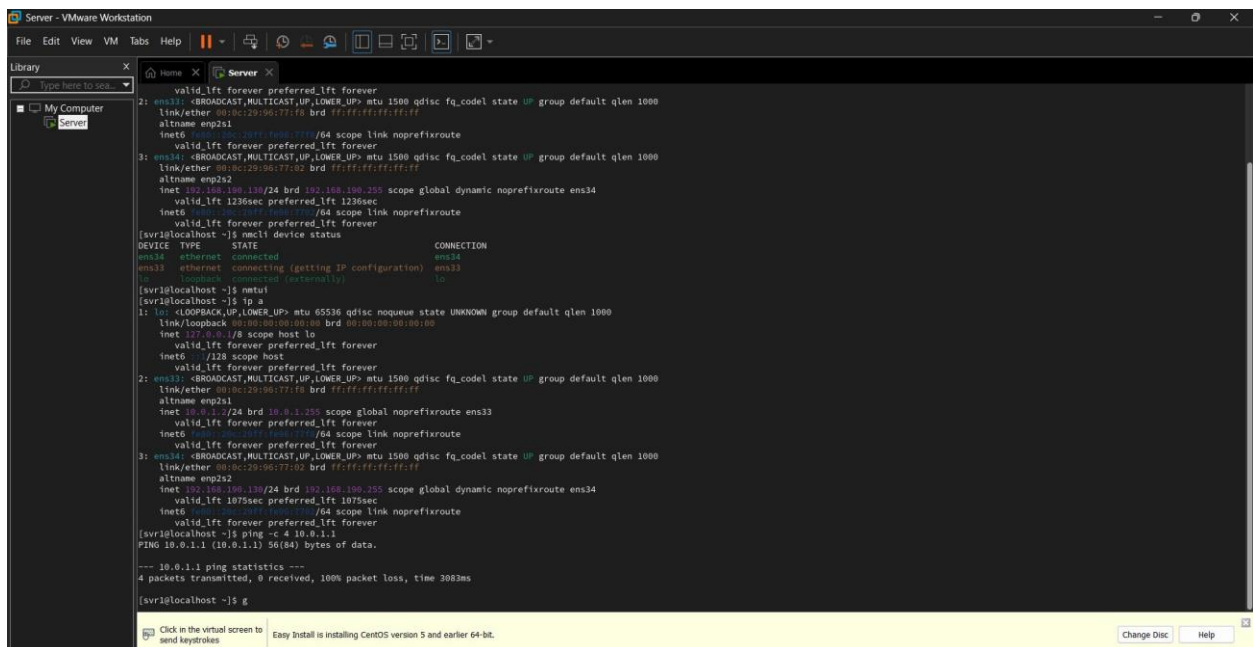
```
Client - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Server
Client

inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
altname enp3s0
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
altname enp1s0
inet 192.168.100.131/24 brd 192.168.100.255 scope global dynamic noprefixroute ens192
valid_lft 1344sec preferred_lft 1344sec
inet6 fe80::473:98d6:37c1:2c82:64 scope link noprefixroute
valid_lft forever preferred_lft forever
client@fedora:~$ nmcli con show
NAME UUID TYPE DEVICE
wired connection 1 3a73b05-9a75-3ae2-b7b7-8ba0148abedf ethernet ens160
wired connection 2 cb77c5da-c313-3625-82f5-0a3d2943dc4 ethernet ens192
lo 926761c1-6060-4000-8000-000000000000 loopback lo
client@fedora:~$ nmcli con mod "wired connection 1" ipv4.method manual ipv4.addresses 10.0.1.3/24 ipv4.gateway 10.0.1.1
client@fedora:~$ [[200-nmcli] con up "wired connection 1"
bash: nmcli: command not found...
nmcli con up "wired connection 1" client@fedora:~$ [[200-nmcli] con up "wired connection 1"
nmcli con up "wired connection 1" client@fedora:~$ [[200-nmcli] con up "wired connection 1"
nmcli con up "wired connection 1" client@fedora:~$ [[200-nmcli] con up "wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/28)
client@fedora:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
altname enp3s0
inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
valid_lft forever preferred_lft forever
inet6 fe80::a3a8:9077:f8a1:2a7a:64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
altname enp1s0
inet 192.168.100.131/24 brd 192.168.100.255 scope global dynamic noprefixroute ens192
valid_lft 1630sec preferred_lft 1630sec
inet6 fe80::473:98d6:37c1:2c82:64 scope link noprefixroute
valid_lft forever preferred_lft forever
client@fedora:~$ g
```

Network Connectivity Testing

Network connectivity between the server and client machines was tested using the ping command.

Successful ping responses were received from both machines, confirming proper network configuration.



```
Server - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Server

valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:96:77:f0 brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 10.0.1.2/24 brd 10.0.1.255 scope link noprefixroute
valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:96:77:f0 brd ff:ff:ff:ff:ff:ff
altname enp2s2
inet 192.168.100.130/24 brd 192.168.100.255 scope global dynamic noprefixroute ens34
valid_lft 1236sec preferred_lft 1236sec
inet6 fe80::a3a8:9077:f8a1:2a7a:64 scope link noprefixroute
valid_lft forever preferred_lft forever
[svr3]localhost ~$ nmcli device status
DEVICE TYPE STATE CONNECTION
ens34 ethernet connected ens34
ens33 ethernet connecting (getting IP configuration) ens33
lo loopback connected (activated) lo
[svr3]localhost ~$ nmcli
[svr3]localhost ~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:96:77:f0 brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 10.0.1.2/24 brd 10.0.1.255 scope global noprefixroute ens33
valid_lft forever preferred_lft forever
inet6 fe80::a3a8:9077:f8a1:2a7a:64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: ens34: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:96:77:f0 brd ff:ff:ff:ff:ff:ff
altname enp2s2
inet 192.168.100.130/24 brd 192.168.100.255 scope global dynamic noprefixroute ens34
valid_lft 1075sec preferred_lft 1075sec
inet6 fe80::a3a8:9077:f8a1:2a7a:64 scope link noprefixroute
valid_lft forever preferred_lft forever
[svr3]localhost ~$ ping -c 4 10.0.1.1
PING 10.0.1.1 (10.0.1.1) 56(84) bytes of data.
--- 10.0.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3083ms
[svr3]localhost ~$ g
```

```
Client - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
Server
Client

Client
$ net 192.168.199.131/24 brd 192.168.199.255 scope global dynamic noprefixroute ens192
$ valid_lft 1344sec preferred_lft 1344sec
$ inet6 fe80::473:98d6:37c1:2c82/64 scope link noprefixroute
$ valid_lft forever preferred_lft forever
Client@fedora:~$ nmcli con show
NAME                UUID                                  TYPE      DEVICE
Wired connection 1  8a780d5-5a75-3aa2-bfb7-bba0148abedf ethernet ens160
Wired connection 2  c577c5da-c313-9a25-82f5-9a3d2943dccc ethernet ens192
Client@fedora:~$ nmcli con mod "Wired connection 1" ipv4.method manual ipv4.addresses 10.0.1.3/24 ipv4.gateway 10.0.1.1
Client@fedora:~$ nmcli con up "[200-nmcli] con up "Wired connection 1"
bash: nmcli: command not found...
nmcli con up "Wired connection 1" Client@fedora:~$ nmcli con up "[200-nmcli] con up "Wired connection 1"
nmcli con up "Wired connection 1" Client@fedora:~$ nmcli con up "[200-nmcli] con up "Wired connection 1"
nmcli con up "Wired connection 1" Client@fedora:~$ nmcli con up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/28)
Client@fedora:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:23:c8:99 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 10.0.1.3/24 brd 10.0.1.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::3d98:9077:f304:1a79/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: ens192: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:23:c8:a3 brd ff:ff:ff:ff:ff:ff
    altname enp1s0
    inet 192.168.199.131/24 brd 192.168.199.255 scope global dynamic noprefixroute ens192
        valid_lft 1638sec preferred_lft 1638sec
    inet6 fe80::473:98d6:37c1:2c82/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
Client@fedora:~$ ping -c 4 10.0.1.2
PING 10.0.1.2 (10.0.1.2): 56(84) bytes of data:
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=5.01 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=4.35 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=1.06 ms
64 bytes from 10.0.1.2: icmp_seq=4 ttl=64 time=3.87 ms

--- 10.0.1.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.050/3.571/5.013/1.507 ms
Client@fedora:~$
```

Conclusion

In summary, this laboratory exercise successfully demonstrated the practical application of Type-2 virtualization. By deploying CentOS and Fedora environments and establishing a dedicated virtual subnet (VMnet2), we validated the process of manual network routing. The final connectivity assessment, confirmed via the Ping command, showed no data loss, proving that the static IP configuration and the virtual bridge were established with high accuracy .